



HANMI POWER

VTU-1100N

MANUAL NO. : VTU-1100N-3.3(2005.03)
(FOR SLIDE VALVE)

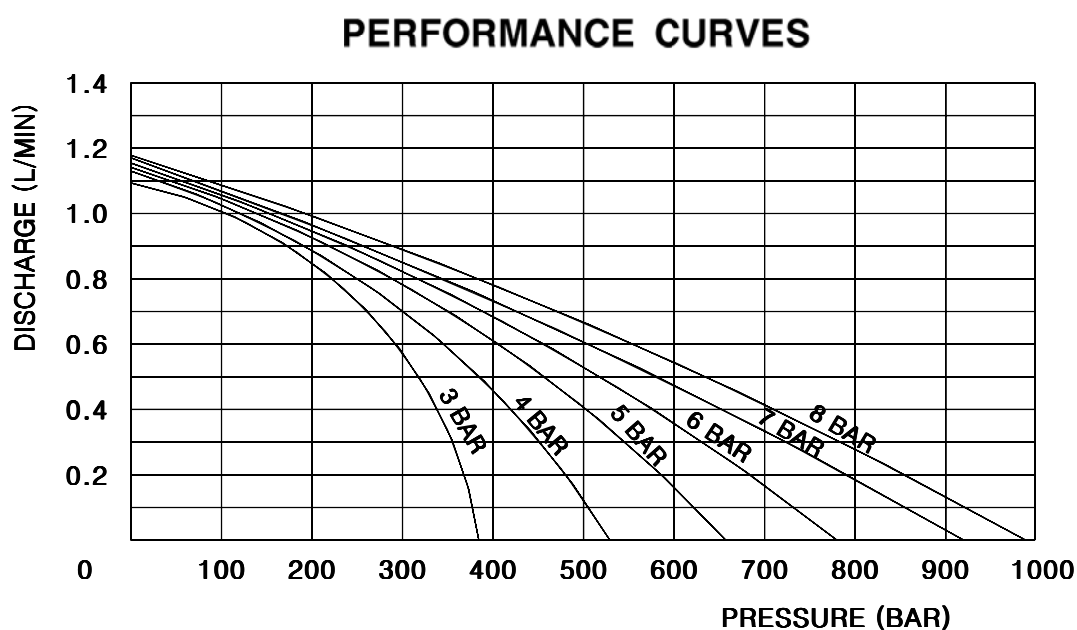
INSTRUCTION MANUAL

FUEL VALVE TEST DEVICE

MODEL : VTU-1100N



HANMI HYDRAULIC MACHINERY CO., LTD.
#700-17, Mora-dong, Sasang-ku, Pusan, Korea
Tel. +82 51 305 2512~4 / Fax. +82 51 301 1721
E-mail : hm131@kornet.net Hm132@kornet.net





B. DESCRIPTION AND OPERATING INSTRUCTION

THE TEST PUMP UNIT TYPE VTU 1100N FOR DIESEL ENGINE FUEL VALVES
CONSISTS OF THE FOLLOWING MAIN SELECTIONS :

- | | | | |
|----------------------------|-------------------|------------|------------------|
| 1) PUMP UNIT : | TYPE VTU 1101.2 : | DRAWING NO | : AP 1100 |
| | | WIDTH | : APPROX 620 mm |
| | | DEPTH | : APPROX 450 mm |
| | | HEIGHT | : APPROX 650 mm |
| | | WEIGHT | : 55 KG |
| | | | |
| 2) TABLE : | TYPE VTU 602-2 : | DRAWING NO | : AP 1104 |
| | | WIDTH | : APPROX 880 mm |
| | | DEPTH | : APPROX 520 mm |
| | | HEIGHT | : APPROX 850 mm |
| | | WEIGHT | : 37KG |
| | | | |
| 3) FIXTURE : | TYPE VTU 603-X : | DRAWING NO | : AP 1105-X |
| | | WIDTH | : APPROX 192 mm |
| | | DEPTH | : APPROX 373 mm |
| | | HEIGHT | : APPROX 785 mm |
| | | WEIGHT | : 10 KG |
| | | | |
| 4) HYDRAULIC HOSE : | TYPE VTU 604-1 : | DRAWING NO | : AP 11 |
| | | LENGTH | : APPROX 1000 mm |
| | | DIAMETER | : APPROX 11.2 mm |
| | | WEIGHT | : APPROX 0.5 KG |



C. OPERATION INSTRUCTIONS

1. BEFORE CONNECTING THE PUMP UNIT

The tank is filled with hydraulic oil (viscosity 7-10Cst at 50°) through the filler cap ⑪ in the right side of the top panel. The tank will hold 6 liters.

Open the oil return valve ② by turning anti-clockwise.

The handle of the regulator-valve ⑬ is closed by turning anti-clockwise.

Compressed air is led through a flexible hose to the stud marked "AIR INLET" ⑨ on the right side of the cabinet. To avoid contamination of the pump, it is essential that an effective air filter and water separator is inserted in the air line. A lubro unit should however be avoided-see lubrication instruction on page 4.

The regulator-valve ③ is opened by turning the handle clockwise until the manometer indicates approx. 5 bar. Pumping of fuel oil through the whole system is now done for about one minute to make sure that no entrapped air is found in the unit. This pumping should only be done, when the oil vessel has run empty. Then the regulator valve ③ is turned anti-clockwise, until manometer ⑥ indicates 0 bar, when the oil return valve ② can be closed.

2. CONNECTION OF THE FUEL VALVE TO THE TEST UNIT

The fuel valve is connected to the pump unit through the stud marked "FUEL OUTLET" ⑩ found on the left side of the cabinet. The fuel valve may be connected by means of a flexible high-pressure-hose.

The test pump is now ready for use.

3. CONTROL PROCEDURE

(NOTICE : This type is not perform to atomizing)

a) LEAK TEST AND VENTING FUNCTION

The handle of the hyd' operating valve ④ is moved to position "OPEN"(4B). The regulator valve ③ is turned slowly clockwise until the manometer ⑦ and the manometer ⑧ indicate approx. 200bar.

The manometer ⑤ will indicate approx. 30~35 bar by pressure reducer control.

Check that there is no leak at the atomizer.

The handle of the hyd' operating valve ④ is then moved to position "CLOSED" (4A) and manometer ⑦ and manometer ⑤ are watched. At approx. 20~25 bar the slide valve of the fuel valve should open, which opening can be seen at manometer ⑤ as an immediate pressure drop.

b) OPENING PRESSURE

The handle of the hyd' operating valve ④ is moved to position "OPEN" (4B).

The regulator valve ③ is turned clockwise until fuel oil is forced through the atomizing holes.

Check that the opening pressure, which is read on manometer ⑦ is correct. please refer to B&W motor instruction - Valve Section.

**c) DIRECTION OF JET**

The regulator valve ③ is turned clockwise until the direction of the jet is clearly visible. The fuel jet should pass through the holes of the jet template. After having finalized the test, the handle of the hyd' operating valve ④ is moved to position "CLOSED" (4A)

d) O-RING SEAL TEST

Oil is pumped through the fuel valve until oil trickles from the venting hole. The venting hole is then closed by means of the plug. The regulator valve ③ is turned clockwise until the manometer ⑦ and the manometer ⑧ indicate 100 bar. The handle of the hyd' operating valve ④ is moved to position (4A). The pressure should remain stable at 100 bar a few minutes. Check that there is no leak at the head of the fuel valve.

e) DISMANTLING OF FUEL VALVE

After this test has been performed the regulator valve ③ is turned anti-clockwise until the manometer ⑥ indicates 0 bar. The valve "OIL RETURN" ② is turned anti-clockwise and the hyd' operating valve ④ is moved to position "OPEN" (4B). The oil pressure has now been completely removed from the pump-system and the fuel valve may be removed.

4. DATA FOR HIGH PRESSURE

The pump unit has a secondary high pressure output of 930 at a primary air pressure of 7 bar and has a maximum output pressure of 1300 bar at an air pressure of abt. 10 bar.

5. CLEANING OF OIL FILTERS

The mesh filter, which is found in the filler cap ⑪ is removed and cleaned when the fuel oil has a tendency of slow flow.

The oil filter ⑫ which is found in the outlet from the oil-vessel is of the "throw-away" type, which should be removed and substituted by a new filter, when the old one has clogged, and the filter should under all circumstances be changed at least once every 6 months.

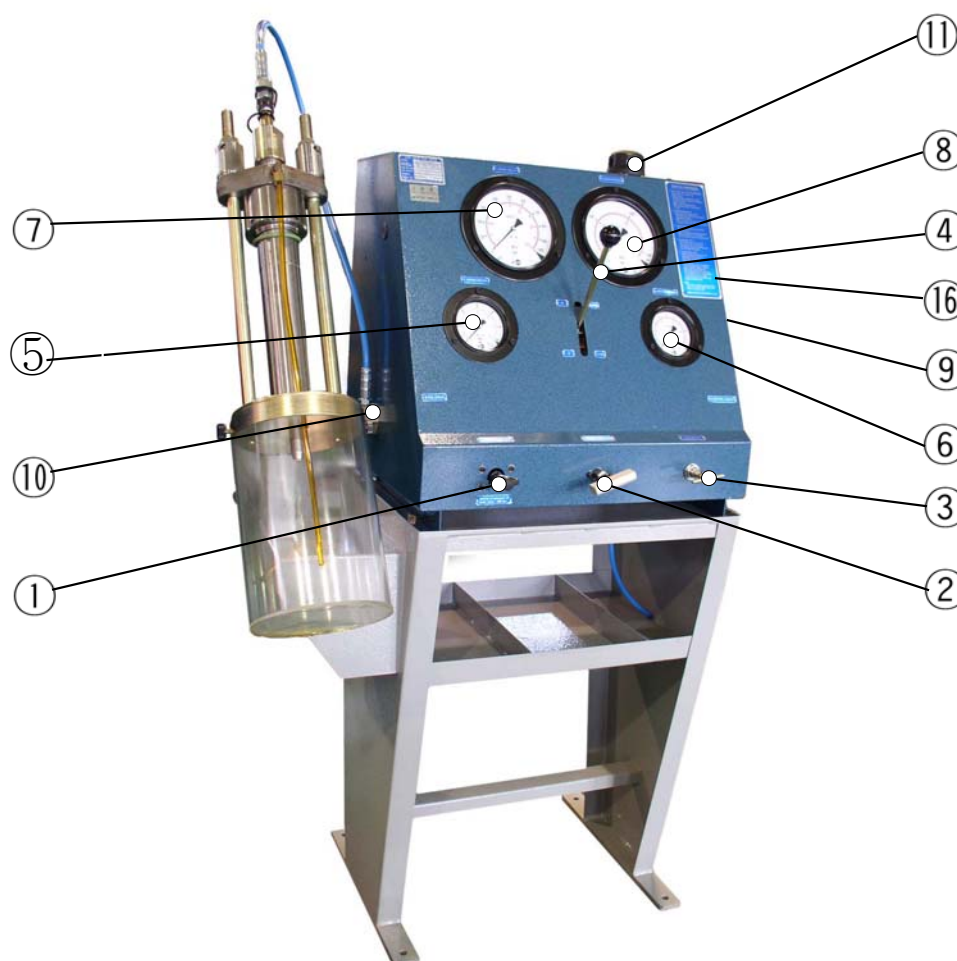
Please note that before removing the filter the oil-vessel should be emptied completely, which is easily done by operating the pump unit without a fuel valve.

6. LUBRICATION INSTRUCTION

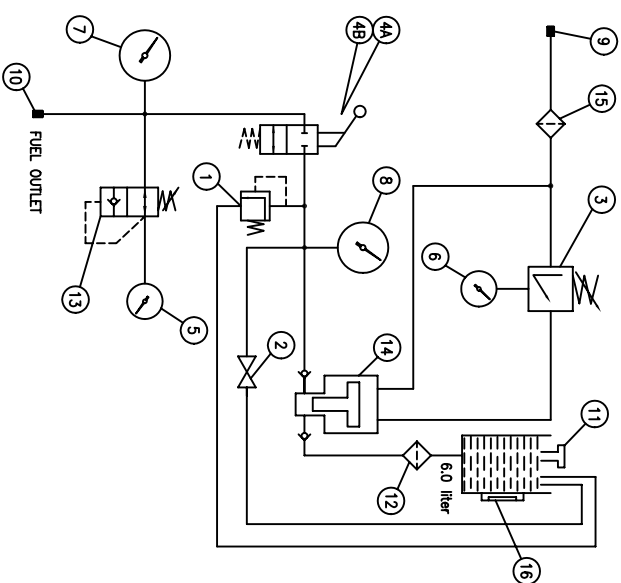
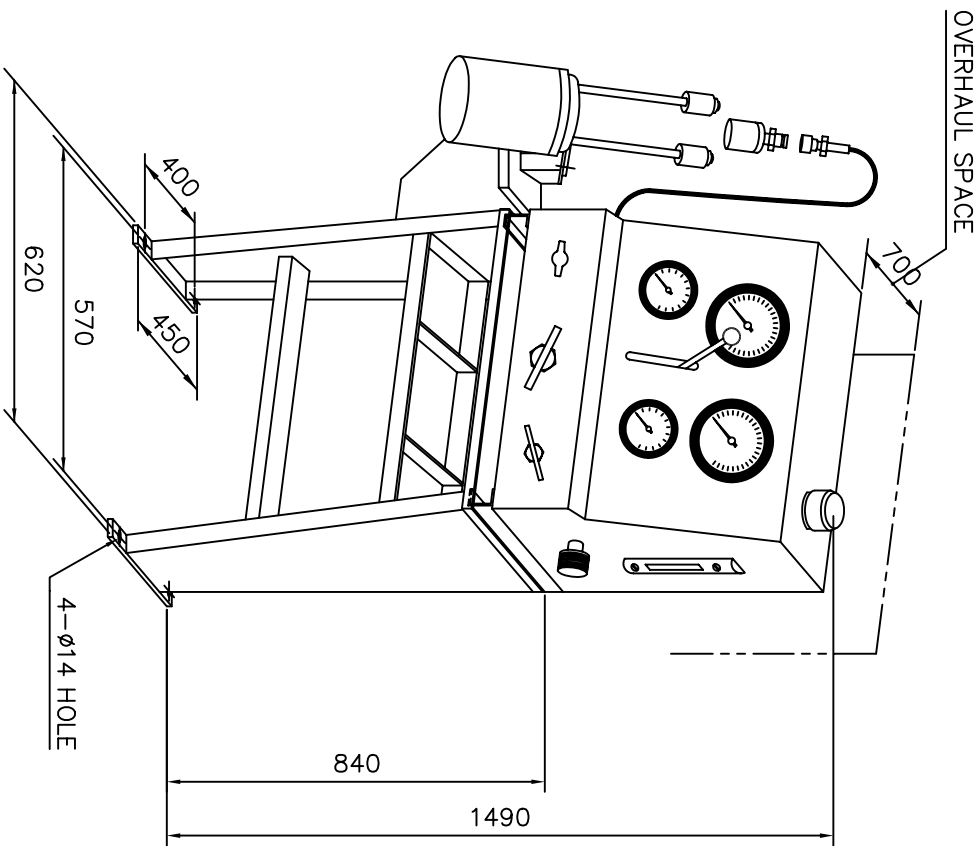
WARNING : AVOID AIRLINE LUBRICATION. The pump has been pre-lubricated and a lubro unit will decrease the efficiency of the pump.

Air valves should be lubricated periodically. Remove exhaust fittings and valve spool and apply a thin coat of the supplied grease.

If primary air drive section is dismantled, always apply a thin coat of the grease to entire inner surface of cylinder tube and plunger o-rings.

**D. OUTLINE OF THE VTU-1100N**

POS.1	PRESSURE RELIEF VALVE	(HM-VTU-6001)	POS.9	STUD " AIR INLET"	(HM-VTU-6009)
POS.2	PRESSURE RELEASE VALVE	(HM-VTU-6002)	POS.10	STUD " FUEL OUTLET"	(HM-VTU-6010)
POS.3	PRESSURE CONTROL VALV	(HM-VTU-6003)	POS.11	FILLER CAP	(HM-VTU-6011)
POS.4	HY'D OPERATING VALVE	(HM-VTU-6004)	POS.12	FILTER	(HM-VTU-6012)
POS.5	PRESSURE GAUGE (0~6Bar/6.0Mpa)	(HM-VTU-6005)	POS.13	PRESSURE REDUCER	(HM-VTU-6013)
POS.6	PRESSURE GAUGE (0~10ar/1.0Mpa)	(HM-VTU-6006)	POS.14	PUMP (inside VTU-1100N unit	(M12-1100-0.0)
POS.7	PRESSURE GAUGE (0~1000Bar/100Mpa)	(HM-VTU-6007)	POS.15	AUTO DRAIN FILTER	(HM-VTU-6015)
POS.8	PRESSURE GAUGE (0~1600Bar/160Mpa)	(HM-VTU-6008)	POS.16	OIL LEVEL GAUGE	(HM-VTU-6016)



1. PRESSURE RELIEF VALVE

2. PRESSURE RELEASE VALVE

3. PRESSURE CONTROL VALVE

4. HYD' OPERATING VALVE

5. PRESSURE GAUGE (0-60 BAR/6.0MPa)

6. PRESSURE GAUGE (0-10 BAR/1.0MPa)

7. PRESSURE GAUGE (0-1000 BAR/100MPa)

8. PRESSURE GAUGE (0-1600 BAR/160MPa)
9. AIR INLET W. CLAW CLUTCH

10. HIGH PRESSURE OUTLET

11. FILTER CAP W. STRAINER

12. OIL FILTER

13. PRESSURE REDUCER

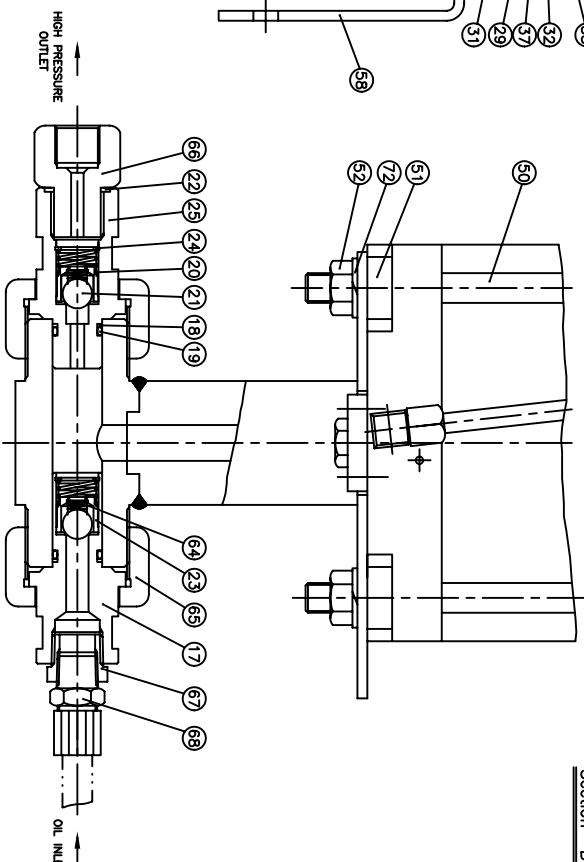
14. PUMP B1100

15. AUTO DRAIN FILTER

16. OIL LEVER GAUGE

Δ	050314	Revision of drawing no.	B.C.KIM		M.W.CHOI
REV.	DATE	REASON FOR MODIFICATION	REV'D	CHK'D	APP'D

APPROVED	SCALE	PROJECT	PROJECTION
M.W.CHOI	N/S	FUEL VALVE TEST DEVICE (YTU-1100N)	
CHECKED	DATE	TITLE	
D.H.KOO	05.3.2	DIMENSION OF OUTLINE & CIRCUIT DIAGRAM	
DESIGNED			
S.Y.CHOI			
HANMI HYDRAULIC MACHINERY CO., LTD.			DRAWING NO.
			M12-6006-0.1

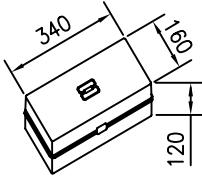
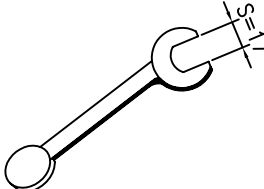
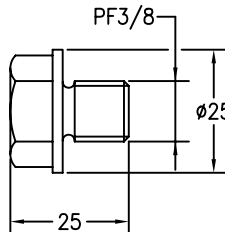
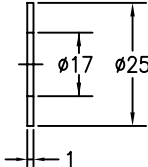
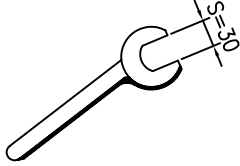
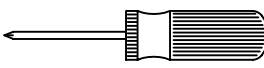
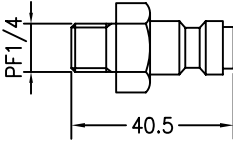
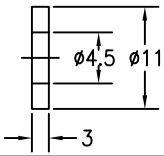


723	WRENCH BOLT	AP1100-23	12-11123-0	3
724	SPRING WASHER	AP1100-24	12-11124-0	4
725	0-RING	AP1100-25	12-11125-0	1
726	0-RING	AP1100-26	12-11126-0	1
727	0-RING	AP1100-27	12-11127-0	1
728	0-RING	AP1100-28	12-11128-0	2
729	0-RING	AP1100-29	12-11129-0	5
730	0-RING	AP1100-30	12-11130-0	1
731	0-RING	AP1100-31	12-11131-0	2
732	0-RING	AP1100-32	12-11132-0	2
733	WASHER	AP1100-33	12-11133-0	1
734	SNAP RING	AP1100-34	12-11134-0	2
735	UPPER VALVE	AP1100-35	12-11135-0	1
736	UPPER VALVE BODY	AP1100-36	12-11136-0	1
737	UPPER VALVE BODY	AP1100-37	12-11137-0	1
738	PLUG	AP1100-38	12-11138-0	1
739	0-RING	AP1100-39	12-11139-0	1
740	SPRIG	AP1100-40	12-11140-0	1
741	SPRIG	AP1100-41	12-11141-0	1
742	SPRIG	AP1100-42	12-11142-0	1
743	SPRIG	AP1100-43	12-11143-0	1
744	SPRIG	AP1100-44	12-11144-0	1
745	SPRIG	AP1100-45	12-11145-0	1
746	SPRIG	AP1100-46	12-11146-0	1
747	SPRIG	AP1100-47	12-11147-0	1
748	SPRIG	AP1100-48	12-11148-0	1
749	SPRIG	AP1100-49	12-11149-0	1
750	SPRIG	AP1100-50	12-11150-0	1
751	SPRIG	AP1100-51	12-11151-0	1
752	SPRIG	AP1100-52	12-11152-0	1
753	SPRIG	AP1100-53	12-11153-0	1
754	SPRIG	AP1100-54	12-11154-0	1
755	SPRIG	AP1100-55	12-11155-0	1
756	SPRIG	AP1100-56	12-11156-0	1
757	SPRIG	AP1100-57	12-11157-0	1
758	SPRIG	AP1100-58	12-11158-0	1
759	SPRIG	AP1100-59	12-11159-0	1
760	SPRIG	AP1100-60	12-11160-0	1
761	SPRIG	AP1100-61	12-11161-0	1
762	SPRIG	AP1100-62	12-11162-0	1
763	SPRIG	AP1100-63	12-11163-0	1
764	SPRIG	AP1100-64	12-11164-0	1
765	SPRIG	AP1100-65	12-11165-0	1
766	SPRIG	AP1100-66	12-11166-0	1
767	SPRIG	AP1100-67	12-11167-0	1
768	SPRIG	AP1100-68	12-11168-0	1
769	SPRIG	AP1100-69	12-11169-0	1
770	SPRIG	AP1100-70	12-11170-0	1
771	SPRIG	AP1100-71	12-11171-0	1
772	SPRIG	AP1100-72	12-11172-0	1
773	SPRIG	AP1100-73	12-11173-0	1
774	SPRIG	AP1100-74	12-11174-0	1
775	SPRIG	AP1100-75	12-11175-0	1
776	SPRIG	AP1100-76	12-11176-0	1
777	SPRIG	AP1100-77	12-11177-0	1
778	SPRIG	AP1100-78	12-11178-0	1
779	SPRIG	AP1100-79	12-11179-0	1
780	SPRIG	AP1100-80	12-11180-0	1
781	SPRIG	AP1100-81	12-11181-0	1
782	SPRIG	AP1100-82	12-11182-0	1
783	SPRIG	AP1100-83	12-11183-0	1
784	SPRIG	AP1100-84	12-11184-0	1
785	SPRIG	AP1100-85	12-11185-0	1
786	SPRIG	AP1100-86	12-11186-0	1
787	SPRIG	AP1100-87	12-11187-0	1
788	SPRIG	AP1100-88	12-11188-0	1
789	SPRIG	AP1100-89	12-11189-0	1
790	SPRIG	AP1100-90	12-11190-0	1
791	SPRIG	AP1100-91	12-11191-0	1
792	SPRIG	AP1100-92	12-11192-0	1
793	SPRIG	AP1100-93	12-11193-0	1
794	SPRIG	AP1100-94	12-11194-0	1
795	SPRIG	AP1100-95	12-11195-0	1

FUEL VALVE TEST DEVICE

TYPE : VTU-1100N

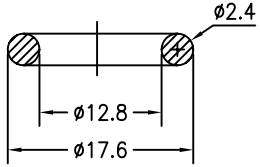
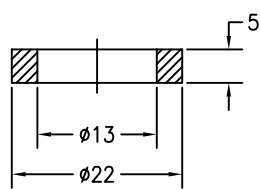
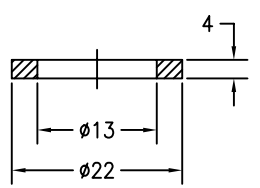
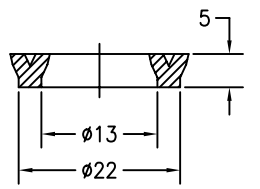
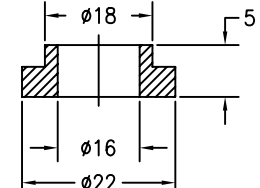
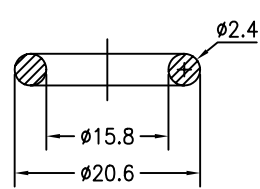
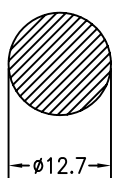
DWG. NO. : M12-1008-0.0

NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SHIP			DRAWING NO.	REMARK
				STAND- ARD	ADDIT- ION	TOTAL		
1	BOX		PLASTIC	1	-	1		
1-1	COMBINATION SPANNER		TOOL STEEL	1	-	1		
1-2	HEXAGON HEAD PLUG WITH FLANGE AND GASKET		STEEL	1	-	1		
				1	-	1		
1-3	SINGLE OPEN ENDED SPANNER		TOOL STEEL	1	-	1		
1-4	+ DRIVER		-	1	-	1		
1-5	MALE COUPLER		STEEL	1	-	1		
1-6	DISC		COPPER	6	-	6		
MANUFACTURER'S NAME		 HANMI HYDRAULIC MACHINERY CO., LTD.						

FUEL VALVE TEST DEVICE

TYPE : VTU-1100N

DWG. NO. : M12-1009-0.0 (1/2)

NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SHIP			OLD DWG NO.	NEW DWG NO.	REMARK
				STAND- ARD	ADDI- TION	TOTAL			
1-7	O-RING		N.B.R	1	-	1	AP1100-09	12-1109-0	
1-8	SPACER		BC3	1	-	1	AP1100-11	12-1111-0	
1-9	BACK-UP RING		TEFLON	1	-	1	AP1100-12	12-1112-0	
1-10	PACKING SEAL		URETANE	1	-	1	AP1100-13	12-1113-0	
1-11	SUPPORT RING		BC3	1	-	1	AP1100-14	12-1114-0	
1-12	O-RING		N.B.R	2	-	2	AP1100-19	12-1119-0	
1-13	BALL		STEEL	2	-	2	AP1100-21	12-1121-0	

MANUFACTURER'S NAME

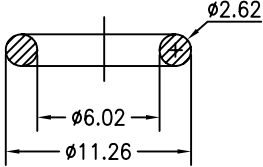
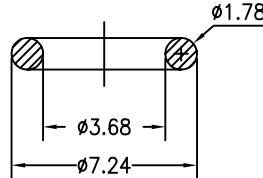
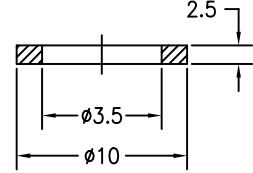
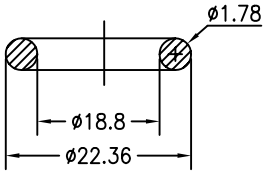
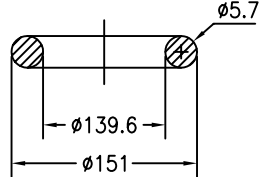
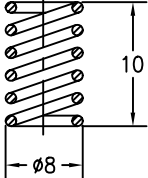


HANMI HYDRAULIC MACHINERY CO., LTD.

FUEL VALVE TEST DEVICE

TYPE : VTU-1100N

DWG. NO. : M12-1009-0.0 (2/2)

NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SHIP			OLD DWG NO.	NEW DWG NO.	REMARK
				STAND- ARD	ADDI- TION	TOTAL			
1-14	O-RING		N.B.R	2	-	2	AP1100-31	12-1131-0	
1-15	O-RING		N.B.R	2	-	2	AP1100-32	12-1132-0	
1-16	WASHER		Bc3	2	-	2	AP1100-33	12-1133-0	
1-17	O-RING		N.B.R	8	-	8	AP1100-45	12-1145-0	
1-18	O-RING		N.B.R	1	-	1	AP1100-48	12-1148-0	
1-19	SPRING		SUP10	2	-	2	AP1100-64	12-1164-0	

MANUFACTURER'S NAME



HANMI HYDRAULIC MACHINERY CO., LTD.